

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	06/09/2026
Team ID	Servicenow
Project Name	Automated Network Request Management
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table 1 & table 2.

Application: Automated Network Request Management

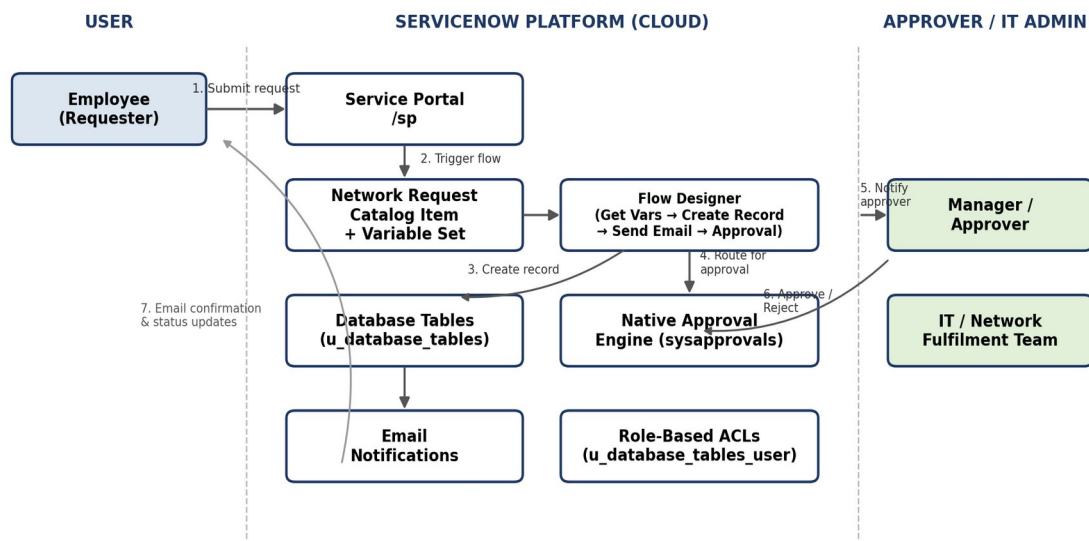


Table-1: Components & Technologies

S.No	Component	Description	Technology
1.	User Interface	How the user interacts with the application	ServiceNow Service Portal (Now Experience UI) — Network Request catalog item
2.	Application Logic-1	Captures request data and creates the request record	Flow Designer — Get Catalog Variables, Create Record
3.	Application Logic-2	Notifies requester of submission	Flow Designer — Send Email action
4.	Application Logic-3	Routes the request for approval and branches on the outcome	Flow Designer — Ask for Approval, If / Else logic
5.	Database	Stores request data and field	Custom ServiceNow table —

S.No	Component	Description	Technology
		configuration	Database Tables (u_database_tables)
6.	Cloud Database	Underlying platform database service	ServiceNow instance database (multi-tenant cloud, vendor-managed)
7.	File Storage	File storage requirements	Not applicable — no file attachments required for this request type
8.	External API	Purpose of external API used in the application	Not applicable — solution uses native ServiceNow capabilities only
9.	Machine Learning Model	Purpose of ML model, if any	Not applicable
10.	Infrastructure (Server / Cloud)	Application deployment	ServiceNow Personal Developer Instance (PDI), hosted on ServiceNow's cloud infrastructure

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Frameworks used in the solution	None — built entirely on the proprietary ServiceNow platform (no custom scripting frameworks)
2.	Security Implementations	Security / access controls implemented	Role-based Access Control Lists (ACLs) scoped to the custom u_database_tables_user role
3.	Scalable Architecture	Justification of scalability	Built on ServiceNow's multi-tenant SaaS architecture; new catalog items and flows can be added without re-architecting
4.	Availability	Justification of application availability	Inherits ServiceNow's platform-level high availability and vendor-managed uptime SLAs
5.	Performance	Design considerations for performance	Native platform indexing and caching; Flow Designer executes asynchronously to avoid blocking user submissions

References:

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>